COMP2300 Set 7.5 Practice Exam Solutions

Part A: Multiple Choice

A1. B

A2. B

A3. B

A4. B

A5. B

A6. C

A7. A

A8. B

A9. C

A10. B

Part B: Computational Problems

B1. Hit and miss rates

(a)

$$\text{Miss Rate} = \frac{40}{500} = 0.08 = 8\%$$

(b)

$$\text{Hit Rate} = 1 - 0.08 = 0.92 = 92\%$$

B2. Address breakdown

(a) Block size is 8 bytes:

$$\log_2 8 = 3$$

So there are 3 offset bits.

(b) Four sets require:

$$\log_2 4 = 2$$

So there are 2 set-index bits.

(c) In a 16-bit address:

$$16 - 3 - 2 = 11$$

So 11 tag bits remain.

B3. Direct-mapped placement

(a)

$$0x34 = 0011\,0100_2$$

Low 5 bits are:

$$10100$$

(b) Offset is the low 3 bits:

$$100_2$$

(c) Set index is the next 2 bits:

$$10_2$$

which is set 2.

B4. Cache benefit calculation

(a) At 25% hit rate:

$$AAT = (0.25)(1) + (0.75)(4) = 0.25 + 3 = 3.25$$

(b) At 50% hit rate:

$$AAT = (0.50)(1) + (0.50)(4) = 0.5 + 2 = 2.5$$

(c) Compare with 3 cycles when disabled:

- 25% hit rate: $3.25 > 3$, so **not beneficial**
- 50% hit rate: $2.5 < 3$, so **beneficial**

B5. SRAM vs DRAM reasoning

(a) SRAM is preferred for caches because it is fast and does not need refresh, which makes it suitable for latency-critical on-chip storage.

(b) DRAM is preferred for large main memory because it has much higher density and lower cost per bit, even though it is slower and requires refresh.

B6. Write-policy comparison

(a) **write-through vs write-back**

- write-through advantage: lower-level memory is always up to date
- write-through tradeoff: more write traffic
- write-back advantage: fewer lower-level writes on repeated updates
- write-back tradeoff: more complexity and dirty-line management

(b) **write-allocate vs no-write-allocate**

- write-allocate advantage: useful if the block will be written/read again soon
- write-allocate tradeoff: may bring in data that is not reused
- no-write-allocate advantage: avoids filling cache on one-off write misses
- no-write-allocate tradeoff: later accesses may miss again

Part C: Past Exam Questions

C1. Cache enable decision

Let hit rate be h. Then:

$$AAT = h(1) + (1-h)(4) = 4 - 3h$$

Cache is beneficial when:

$$4 - 3h < 3$$

$$-3h < -1$$

$$h > \frac{1}{3}$$

So the cache is beneficial when hit rate is greater than $33.\overline{3}\%$.

C2. Levels in the memory hierarchy

One reasonable ordering is:

$$\text{Registers} \rightarrow \text{L1} \rightarrow \text{L2} \rightarrow \text{L3} \rightarrow \text{DRAM} \rightarrow \text{Disk/Flash}$$

Lower levels are larger but slower because technology optimized for density and low cost per bit generally has longer access latency and is further from the CPU.

C3. Cache misses in pipelined execution

Long-latency cache misses create large stalls if the processor can only wait in order. Later out-of-order designs try to overlap other independent instructions and even multiple memory requests while a miss is being serviced, which is why memory-level parallelism becomes valuable once cache-miss latency is significant.